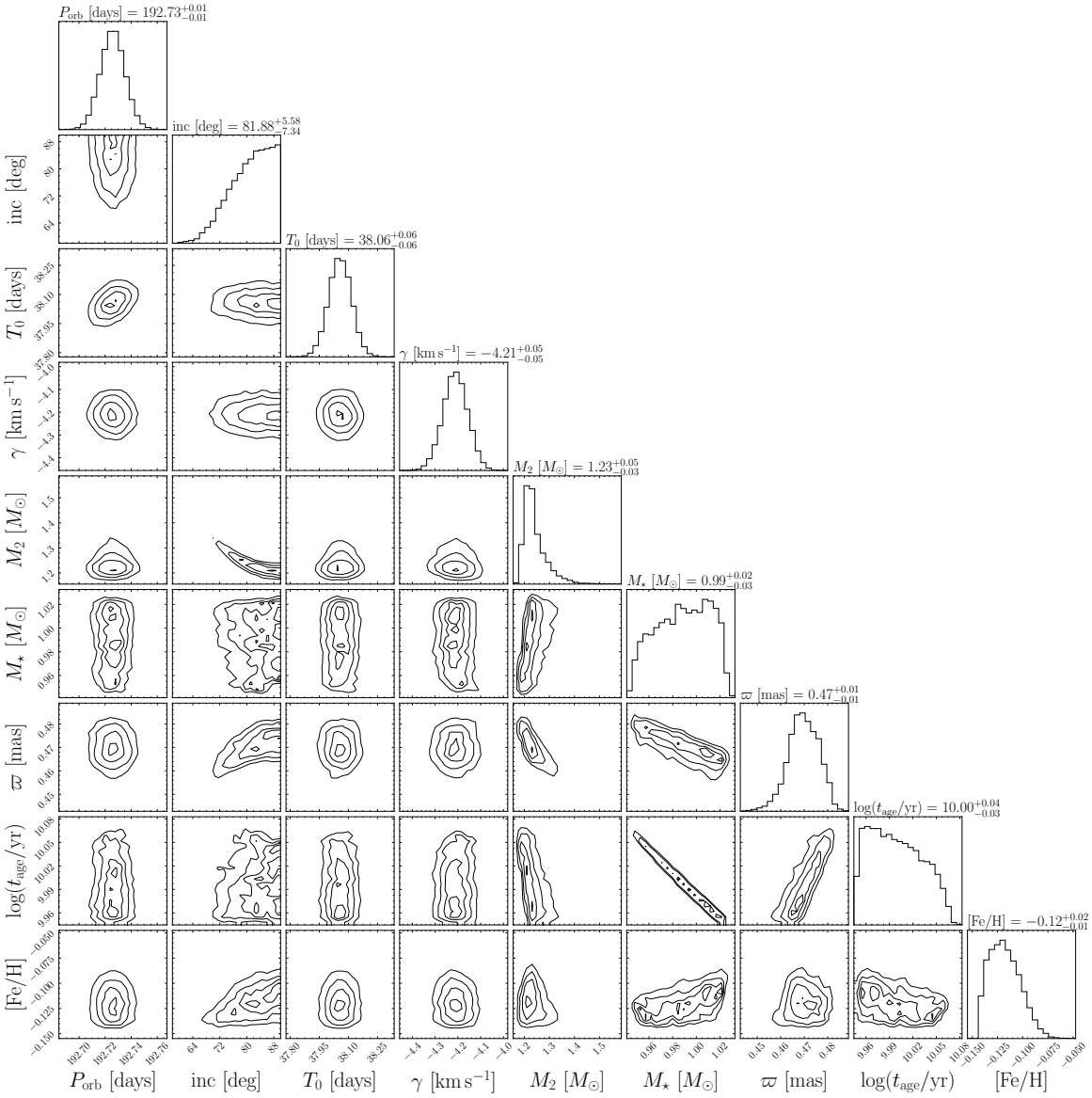

Figure 7. Corner plot for orbital and stellar parameters of IGR J16194-2810, assuming an ellipsoidal light curve and two-body Keplerian orbit. The diagonal entries display the marginal distribution of each parameter, while each of the other panels displays a joint distribution. We display the distributions for the present-day mass and surface metallicity (derived using stellar models), as opposed to the initial mass and metallicity (which are directly sampled). The epoch of conjunction T_0 is provided relative to JD $= 2460000$.

cretion flow relative to a red giant — we do not expect to see eclipses in SyXBs in the optical bands. Indeed, our joint fit implies $i \gtrsim 75^\circ$ and thus predicts an X-ray eclipse. Future targeted observations of IGR J16194-2810 at superior conjunction can test this prediction.

We use truncated uniform distributions to enforce the orbital inclination to lie in $[0^\circ, 90^\circ]$ and restrict the epoch of conjunction to fall within a time range spanning less than one orbital period. We also place a Gaussian prior on the parallax to IGR J16194-2810 based on the *Gaia* DR3 value after correcting for the parallax zeropoint (Lindegren et al. 2021).

We use `isochrones` (Morton 2015) to interpolate on MIST stellar models (Choi et al. 2016) to predict the red giant's radius, effective temperature, surface gravity, present-day mass, and present-day surface metallicity from its age, initial mass, and initial metallicity. We use a truncated uniform prior to restrict the age to $\leq$ 13.5 Gyr. In red giants, the present-day surface metallicity tends to be higher than the star's initial metallicity by ~ 0.04 dex due to dredge-up (Dotter et al. 2017).

Assuming Gaussian uncertainties, we add a term $L_{\rm SPEC}$ to the likelihood to compare the predicted stellar parameters against those derived from the comparison with the GALAH spectra (Section 5.4). Since `isochrones` predicts a value for the radius of the red giant, this term also includes a com-

Table 4. Derived orbital and stellar parameters of IGR J16194-2810. Errors on the median constraints are derived from the 16th and 84th percentiles. Quantities below the dividing line are predicted using the `isochrones` framework and not directly sampled using MCMC techniques.

Parameter	Description	Median Constraint (This Work)	MAP Constraint (This Work)
(1)	(2)	(3)	(4)
P	Orbital Period	192.73 ± 0.01 days	192.73 days
i	Inclination	$(82^{+6}_{-7})^\circ$	85.5°
T_0	Epoch of Conjunction (JD -2460000)	38.06 ± 0.06	38.06
γ	Center-of-Mass RV	-4.21 ± 0.05 km s^{-1}	-4.217 km s^{-1}
$M_{\rm NS}$	Neutron Star Mass	$1.23^{+0.05}_{-0.03}\,M_\odot$	$1.223\,M_\odot$
ϖ	Parallax	$0.470^{+0.007}_{-0.006}$ mas	0.469 mas
$\log\left(t_{\rm age}/{\rm yr}\right)$	Red Giant Age	$10.00^{+0.04}_{-0.03}$	9.9737
M_*	Red Giant Mass	$0.99^{+0.02}_{-0.03}\,M_\odot$	$1.0084\,M_\odot$
R_*	Red Giant Radius	$54.8 \pm 0.7\,R_\odot$	$54.90\,R_\odot$
L_*	Red Giant Luminosity	$518^{+13}_{-15}\,L_\odot$	$520\,L_\odot$
$T_{\rm eff}$	Effective Temperature	3723^{+6}_{-7} K	3722 K
$\log\left(g/{\rm cm\ s^{-2}}\right)$	Surface Gravity	$0.958^{+0.004}_{-0.009}$	0.963
[Fe/H]	Present-Day Surface Metallicity	$-0.12^{+0.02}_{-0.01}$	-0.1026
$v \sin i$	Projected Equatorial Rotational Velocity	14.1 ± 0.2 km s^{-1}	14.37 km s^{-1}

parison of the predicted $v \sin i = 2\pi R \sin i / P$ with the empirical value from Section 5.5. In other words:

$$\ln L_{\rm SPEC} = \sum_{X \in \{T_{\rm eff},\, \log g,\, v \sin i\}} -\frac{1}{2}\left(\frac{X - \langle X \rangle}{\sigma_X}\right)^2 \quad (6)$$

where $\langle X \rangle$ is the best-fit value for parameter X and σ_X is the corresponding uncertainty. We do not include the [Fe/H] derived from comparison with the GALAH spectra in this likelihood term due to the reasons discussed in Section 5.4.

We jointly fit all of the available data, using the following total log likelihood:

$$\ln L = \ln L_{\rm LC} + \ln L_{\rm RV} + \ln L_{\rm SED} + \ln L_{\rm SPEC} \quad (7)$$

We use ensemble MCMC sampling (`emcee`; Foreman-Mackey et al. 2013) with 64 walkers and 5000000 total iterations to derive best-fit orbital and stellar parameters for IGR J16194-2810. We present the resulting corner plot in Figure 7, and report the median and maximum a posteriori (MAP) constraints in Table 4. We derive a best-fit NS mass of $1.23^{+0.05}_{-0.03}\,M_\odot$. Our best-fit orbital period is 192.73 ± 0.01 d, in agreement with our Lomb-Scargle analysis. We find an orbit that is close to edge-on, with a red giant of mass $0.99^{+0.02}_{-0.03}\,M_\odot$ that is $\approx 82\%$ Roche lobe-filling. The derived effective temperature of $T_{\rm eff} = 3723^{+6}_{-7}$ K and surface gravity of $\log\left(g/{\rm cm\ s^{-2}}\right) = 0.958^{+0.004}_{-0.009}$ agree with both the GALAH and SED-inferred values to within $\approx 2\sigma$. The best-fit present-day surface metallicity of $[{\rm Fe/H}] = -0.12^{+0.02}_{-0.01}$ falls in-between the GALAH and SED-inferred values, as expected. For a red giant age of 9.9 ± 0.8 Gyr, the corresponding initial mass and initial metallicity are $1.00^{+0.02}_{-0.03}\,M_\odot$ and $-0.16^{+0.02}_{-0.01}$, respectively. Finally, the predicted projected equatorial rotational velocity of $v \sin i = 14.1 \pm 0.2$ km s^{-1} is consistent within 2σ to the value derived from comparison with the GALAH spectra (see Section 5.5). We note that the uncertainties on the predicted stellar parameters are implausibly small, and are likely to be dominated by unaccounted-for systematics in the stellar models. Nevertheless, most of our results are in good agreement with those derived in the contemporaneous analysis of Hinkle et al. (2024).

5.7. *Spin Period Analysis*

Luna (2023) reported a measurement of the NS spin period from TESS data. We attempt to reproduce this measurement and assess how robust it is here. We download high-quality (i.e. `quality_mask='hard'`) TESS data for IGR J16194-2810 from Sectors 12 and 39 using the Quick Look Pipeline (Huang et al. 2020). After removing outliers and de-trending the light curves using a Savitzky-Golay filter, we perform separate Lomb-Scargle periodogram analyses for the entire dataset and over each of the three temporal regions delineated by Luna (2023).

We show the normalized TESS light curves for each temporal region and for the entire dataset in the right column of Figure 8. In each temporal region, we fold the light curve according to the best-fit period derived from the corresponding Lomb-Scargle periodogram, presented in the left column of

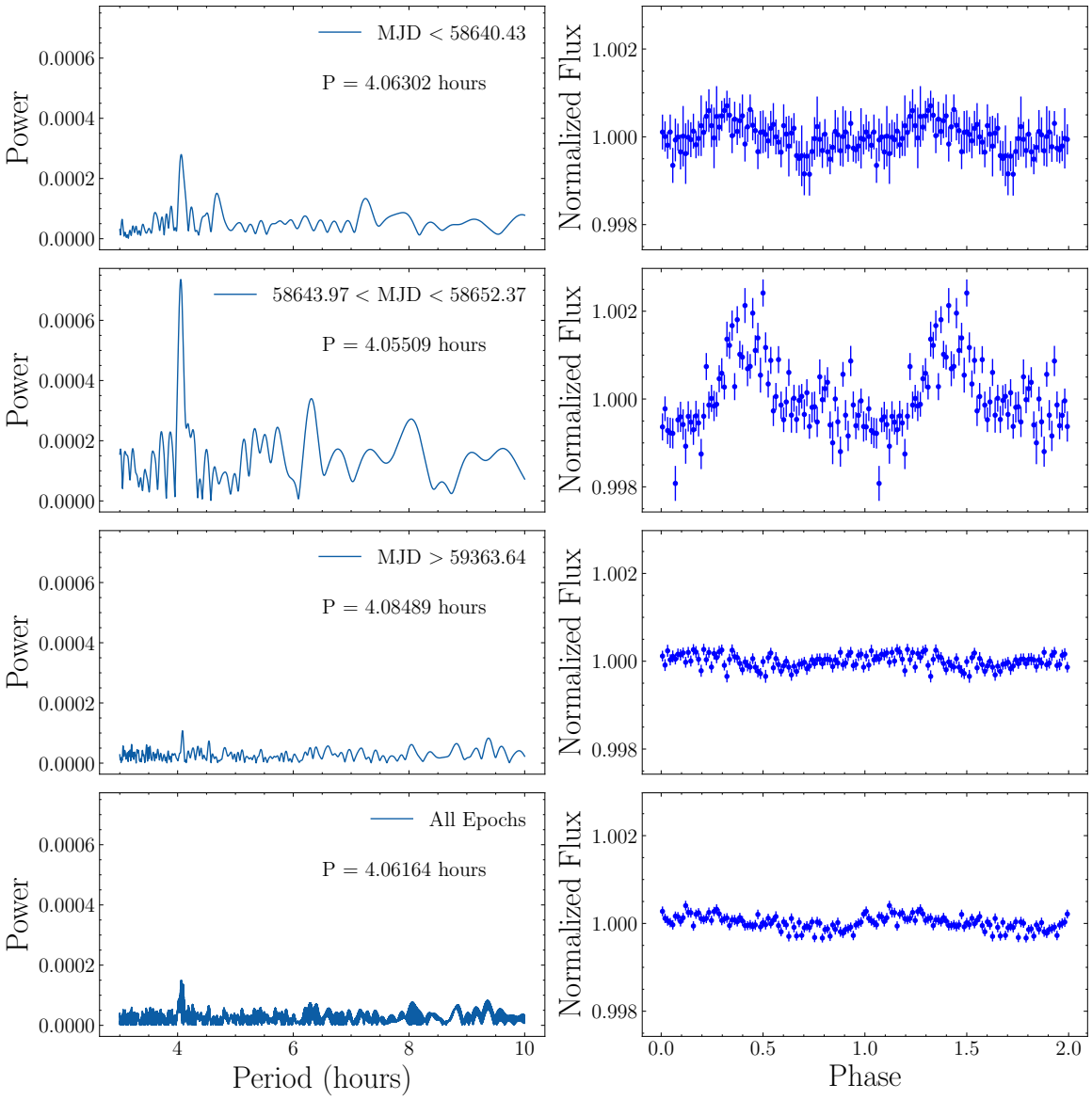

Figure 8. Left Column: Lomb-Scargle periodograms for different temporal regions of TESS photometric data. The periodogram is normalized such that the best-fit sinusoid at each frequency has an amplitude of the corresponding power. Right Column: Normalized TESS light curves folded on the corresponding best-fit periods and binned at 80 bins/cycle. The strongest signal ($P_{\rm spin} = 4.05509$ hr) is detected in $58643.97 <$ MJD < 58652.37, where the peak-to-peak variation is about 0.3% of the red giant luminosity.

Figure 8. Each light curve is then binned at 80 bins/cycle. In the first two regions (i.e. the first and second halves of Sector 12), we identify a peak at a period of ~ 4 hours. Hints of a weak peak are also detectable in the third region and in the full light curve. While this is suggestive of a NS spin period on this timescale, the weakness and transience of the signal, combined with the lack of a convincing visual pattern in the folded light curve, suggest that the spin period measurement should not be considered to be definitive. Continuous X-ray observations of IGR J16194-2810 over 12+ hours would be useful to confirm the period detection.

The shape of our folded and binned TESS light curve from the second half of sector 12 does not match that provided by Luna (2023). The folded light curve shown in his Figure 4 appears significantly less noisy and more coherent than the one yielded by our analysis. We were unable to determine the reason for this discrepancy (Luna, private communication).

The strongest signal is detected in the second half of Sector 12 ($58643.97 <$ MJD < 58652.37). In that temporal region, we find the best-fit spin period to be $P_{\rm spin} = 4.06$ hr, with a variation in optical luminosity over one spin period of $\sim 1.5\,L_{\odot}$. The amplitude of this variation is comparable to the X-ray luminosity observed from the X-ray source in IGR J16194-2810 (see Table 2). While this emission is typical for neutron star accretion flows, it could also arise from an ac-